

$$= 0.0266 \frac{Bq}{r} = 0.0266 \text{ r}$$

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(a)(i) It means direction of force on a (small test) mass/ direction of acceleration of a (small test) mass.

(ii) It is because field lines are radial. The lines near the surface are (approximately) parallel. With parallel lines, the field strength is constant, thus constant acceleration of free fall.

(b)(i) GPE at infinity is maximum at 0J and KE is minimum at 0J.

By conservation of energy,

Gain in GPE from surface to infinity = Lost in KE from surface to infinity

$$\frac{GMm}{r} = \frac{1}{2} mv^2 \quad \text{where mass is the mass of one gas molecule}$$

$$v = \sqrt{\frac{2GM}{r}} \quad (\text{shown})$$

$$(ii) \text{ The escape velocity, } v = (2 \times 6.67 \times 10^{-11} \times 0.131 \times 10^{23} / (1.19 \times 10^6))^{1/2} \\ = 1212 = 1200 \text{ ms}^{-1}$$

(iii) Mercury has a higher escape velocity than Pluto which is 4248 ms^{-1} .

Mercury is closer to sun and Mercury is hotter.

Molecules on Mercury have speed higher than the escape speed, thus they are fast enough to escape from Mercury. Mercury has no atmosphere.

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